Coastal dunes

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1 Overview

Coastal foredunes grow where wind blows sand off the beach and dune plants trap it. Teixeira et al. (2023) study this with DuBeVeg, a cellular automaton in the family of Werner's sand-slab model: the surface is a stack of slabs on each cell, slabs are picked up and hop downwind until they land, avalanches keep slopes below the angle of repose, the sea resets the beach below the water level, and two kinds of plant grow or die back with the sand they gain. In the original model a slab hops one cell; the authors make it hop a saltation length of 5 cells, with slabs five times thinner so that the potential transport is the same, and let vegetation trap slabs part-way through a hop. On a wide beach this changes how the dune grows.

This implementation is an adaptation: the authors' code is not published, the domain is narrower, and several constants come from the base model's documentation.

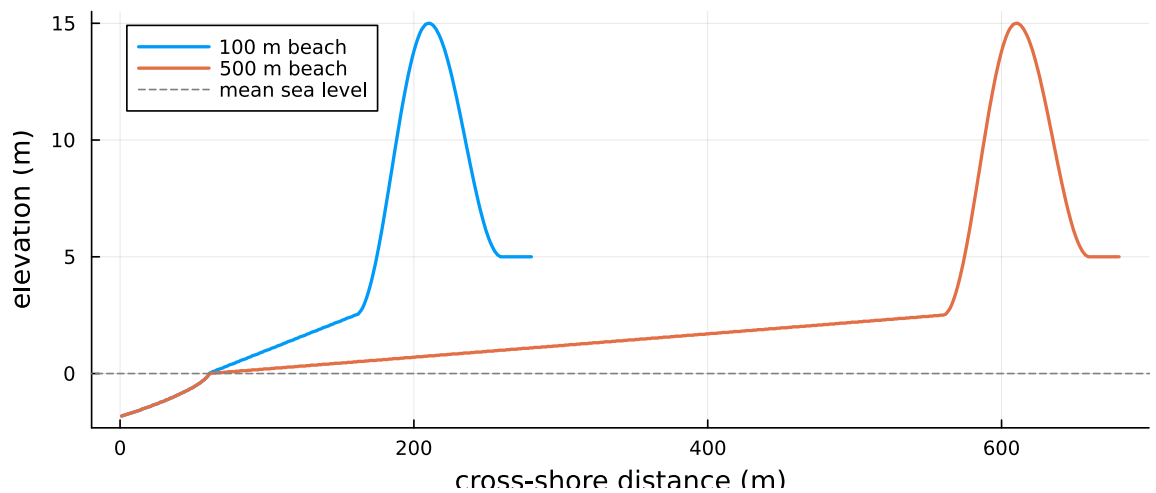
```
ENV["GKSwstype"] = "100"
using EcologicalCellularAutomata
using CellularAutomataCore
using Plots
using Random
using Statistics
```

2 The model

Each aeolian iteration (50 a year), every bare, dry cell outside the shadow of higher ground gives up a slab with probability `(1 - ρ) 0.5`, where ρ is its plant cover. A slab moves landward and, at the end of each hop of `L` cells, lands with probability `ρ (1 - 0.1) + 0.1` (always in the shadow); with `L = 5`, it can also land part-way through a hop on cells whose plant cover is at least 0.3. Every second iteration the sea, with run-up from waves, moves the cells it reaches towards the reference beach, removing or adding each slab between them with the same probability. Once a year the plants grow or die back with the sand each cell gained, each species spreading from neighbours of its own, and the year's sea thins them.

The beach rises from the shoreline to the foot of a foredune 15 m high and 100 m wide; the source compares a narrow beach of 100 m with a wide one of 500 m.

```
plot(; xlabel="cross-shore distance (m)", ylabel="elevation (m)", size=(700, 300),
     legend=:topleft)
for (beach, c) in ((100, 1), (500, 2))
    prof = dune_profile(; beach, h=0.02)
    plot!(prof.elevation .* 0.02; lw=2, color=c, label="$beach m beach")
end
hline!([0]; color=:gray, ls=:dash, label="mean sea level")
```

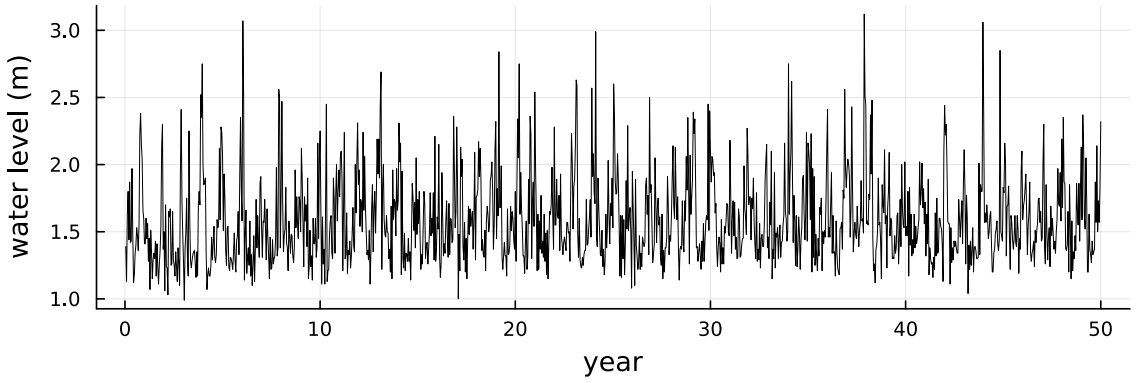


2.1 Water levels

The source forces the sea with the highest water level in each spring-neap cycle at Scheveningen, 1970 to 2020, which it plots (its Fig. 5d) but does not tabulate. The measurements are public, from Rijkswaterstaat, and `scheveningen_water_levels` holds them: the highest level in each block of 14.6 days, 25 a year. They agree with the figure, whose median is about 1.5 m, with a tenth of the cycles above about 2 m and a highest level of about 3.2 m, and they run higher in winter than in summer. Wave run-up adds a few tenths of a metre more, and the level the sea

reaches most years keeps the beach below it bare. The levels are against the Dutch datum, NAP, where the source writes mean sea level; no correction is made. `problem(:dune_formation; levels=:synthetic)` draws a series with much the same distribution from the problem's seed, 1.43 m plus a Gumbel-distributed surge of scale 0.24 m (`dune_water_levels`).

```
levels = scheveningen_water_levels()
plot((1:length(levels)) ./ 25, levels; lw=0.6, color=:black, legend=false,
     xlabel="year", ylabel="water level (m)", size=(700, 240),
left_margin=4Plots.mm,
     bottom_margin=4Plots.mm)
```




```
in_year(months...) = [l for (k, l) in enumerate(levels) if mod1(k, 25) in months]
(median=round(median(levels); digits=2), q90=round(quantile(levels, 0.9); digits=2),
 highest=round(maximum(levels); digits=2),
 winter=round(mean(in_year(23, 24, 25, 1, 2, 3)); digits=2),
 summer=round(mean(in_year(9:15...)); digits=2))
```

```
(median = 1.52, q90 = 2.04, highest = 3.12, winter = 1.79, summer = 1.38)
```

2.2 Profiles from pieces

A profile is built by joining pieces from the sea to the land: a Dean shoreface, a straight beach, the dune ridge and the plateau behind it. A piece runs from one elevation to another, and `*` joins two pieces only where the first ends at the elevation where the second starts. Joining is associative and an empty piece changes nothing, so profiles compose like paths between elevations (morphisms in a category whose objects are elevations), and a scenario is a composite:

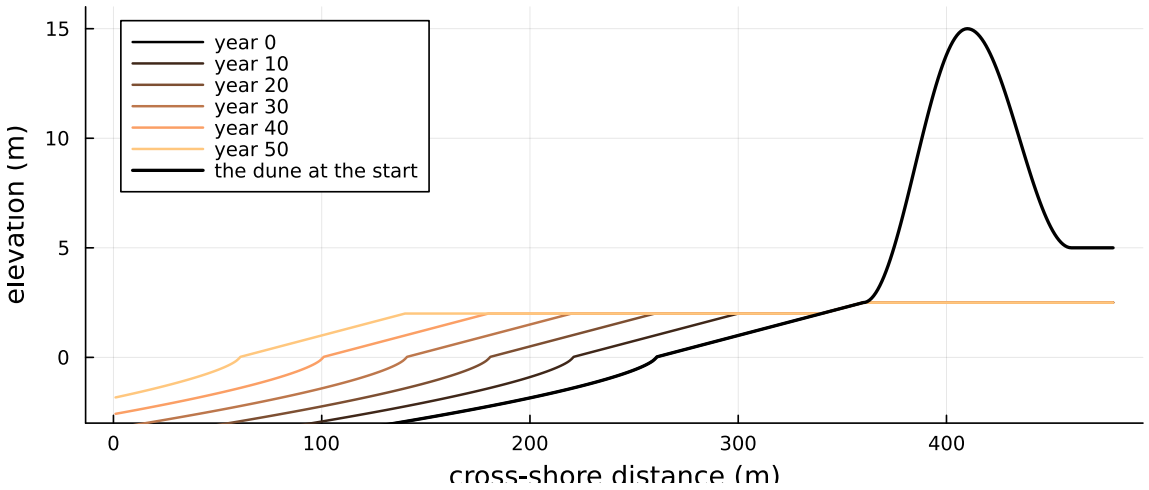
```
narrow = shoreface_segment(60) * slope_segment(100, 0, 2.5) *
         ridge_segment(100, 2.5, 15, 5) * flat_segment(20, 5; kind=:hinterland)
(narrow, profile_heights(narrow, 280; h=0.02) == dune_profile(;
beach=100).elevation)
```

```
(ProfileSegment(shoreface * beach * dune * hinterland: -1.8391426377445272 → 5.0 m
over 280 m), true)
```

The reference surface the sea works towards is the same profile with the dune and the land behind it levelled at the dune's foot. `reference_profile` does this piece by piece, so it respects joins: it maps composites to composites.

Teixeira et al.'s third scenario is a prograding beach, which builds out 4 m a year, from 100 m to 300 m wide over 50 years. In their Fig. 2b the profile below a berm at 2 m moves seaward and the berm lengthens. With pieces, that is one more piece: the beach is cut where it reaches 2 m and a level berm of `4t` m is joined in. Level pieces at one elevation add, a berm of `a` m joined to one of `b` m being a berm of `a + b` m, so building out is time acting on the profile by joining berms. `problem(:dune_formation; scenario=:prograding)` puts the narrow beach on a domain 200 m longer out to sea, anchored at the land, and replaces the reference surface with the year's at the end of every year; the sea, resetting cells towards it, builds the beach out.

```
plot(; xlabel="cross-shore distance (m)", ylabel="elevation (m)", size=(700, 300),
     legend=:topleft, ylims=(-3, 16))
for y in 0:10:50
    ref = reference_profile(dune_landform(; beach=100, offshore=260, berm=4y))
    plot!(profile_heights(ref, 480); lw=1.5, color=cgrad(:copper)[y / 50],
    label="year $y")
end
plot!(profile_heights(dune_landform(; beach=100, offshore=260), 480); color=:black,
lw=2,
      label="the dune at the start")
```



3 The same potential transport

The source's Eq. 3 gives the potential transport `Q = h L (pe0 / pd0) n`: with slabs of 0.1 m hopping 1 cell or of 0.02 m hopping 5, 25 m^3 per metre of beach a year. On a bare, dry beach the slabs crossing a line do carry that:

```
function flat_flux(h, L; rows=40, iterations=20)
    p = merge(defaults(build(:dune_formation)),
              (; h, L, rho_star=Inf, hydro_every=10^6, iterations=10^6))
    total = 0
    for t in 1:iterations
        u = (topo=fill(Int32(round(2.9 / h)), rows, 400), pioneer=zeros(rows, 400),
             stabilizer=zeros(rows, 400), balance=zeros(Int32, rows, 400),
             zref=zeros(Int32, rows, 400), sea=zeros(rows, 400))
        u.topo[:, 200:end] .= round(Int32, 3.1 / h)
        total += dune_step!(u, p, t, Random.Xoshiro(t))
    end
    return round(total * h * 50 / (rows * iterations); digits=1)
end
(hop_1=flat_flux(0.1, 1), hop_5=flat_flux(0.02, 5))
```

```
(hop_1 = 24.6, hop_5 = 24.9)
```

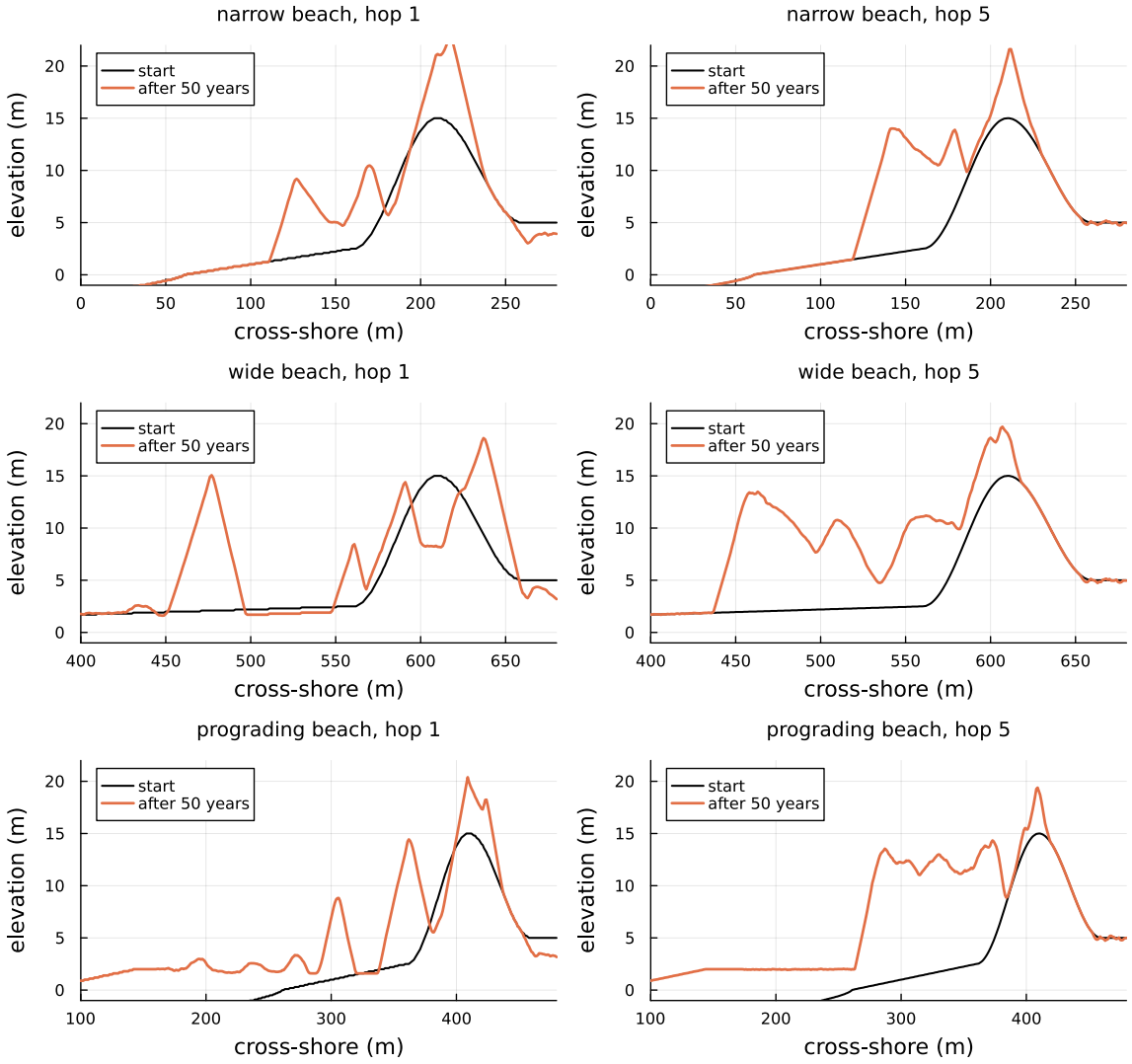
4 Fifty years of dune growth

One run of 50 years for each beach and representation, on 20 rows of 1 m (the source runs 400 rows five times). We record the sand crossing the dune toe, the first cell above 3 m, each year, and the dune's volume and width every ten years.

```
function dune_run(scenario, representation; years=50, rows=20, seed=1)
    prob = problem(:dune_formation; scenario, representation, alongshore=rows,
years, seed)
    p, u, h = prob.p, deepcopy(prob.u0), prob.p.h
    flux = zeros(years)
    measures = [dune_measures(u; h)]
    profiles = [vec(mean(u.topo; dims=1)) .* h]
    for t in 1:(50years)
        crossed = dune_step!(u, p, t, Random.Xoshiro(hash((prob.seed, t, :dune))))
        flux[cld(t, 50)] += crossed * h / rows
        if t % 500 == 0
            push!(measures, dune_measures(u; h))
            push!(profiles, vec(mean(u.topo; dims=1)) .* h)
        end
    end
    return (; flux, measures, profiles, final=u.topo .* h)
end
cases = [(s, r) for s in (:narrow, :wide, :prograding) for r in (:default,
:saltation)]
runs = Dict(zip(cases, fetch.([Threads.@spawn dune_run(c...) for c in cases])))
nothing
```

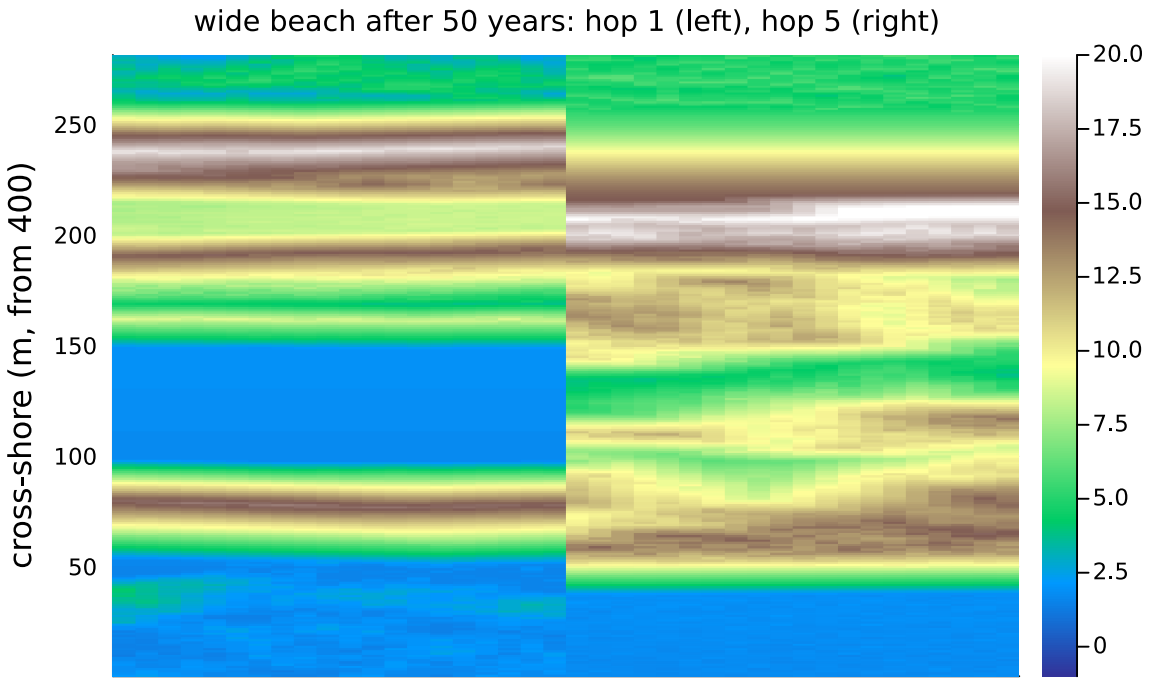
The same runs drive the integrator: `dune_step!` with the generator seeded from the problem's seed and the iteration is what `solve` runs at each step.

```
plot([begin
          r = runs[(s, rep)]
          plot(r.profiles[1]; color=:black, lw=1.5, label="start",
               title="$s beach, hop $(rep === :saltation ? 5 : 1)",
titlefontsize=10,
               xlabel="cross-shore (m)", ylabel="elevation (m)", legend=:topleft,
               xlims=s === :wide ? (400, 680) : s === :prograding ? (100, 480) : (0,
280),
               ylims=(-1, 22))
          plot!(r.profiles[end]; color=2, lw=2, label="after 50 years")
      end
      for (s, rep) in cases]...; layout=(3, 2), size=(850, 800))
```




```
heatmap(hcat(permutedims(runs[(:wide, :default)].final[:, 400:end]),
             permutedims(runs[(:wide, :saltation)].final[:, 400:end])); c=:terrain,
```

```
        clims=(-1, 20), xticks=false, ylabel="cross-shore (m, from 400)", size=(600,
360),
        title="wide beach after 50 years: hop 1 (left), hop 5 (right)",
titlefontsize=10)
```



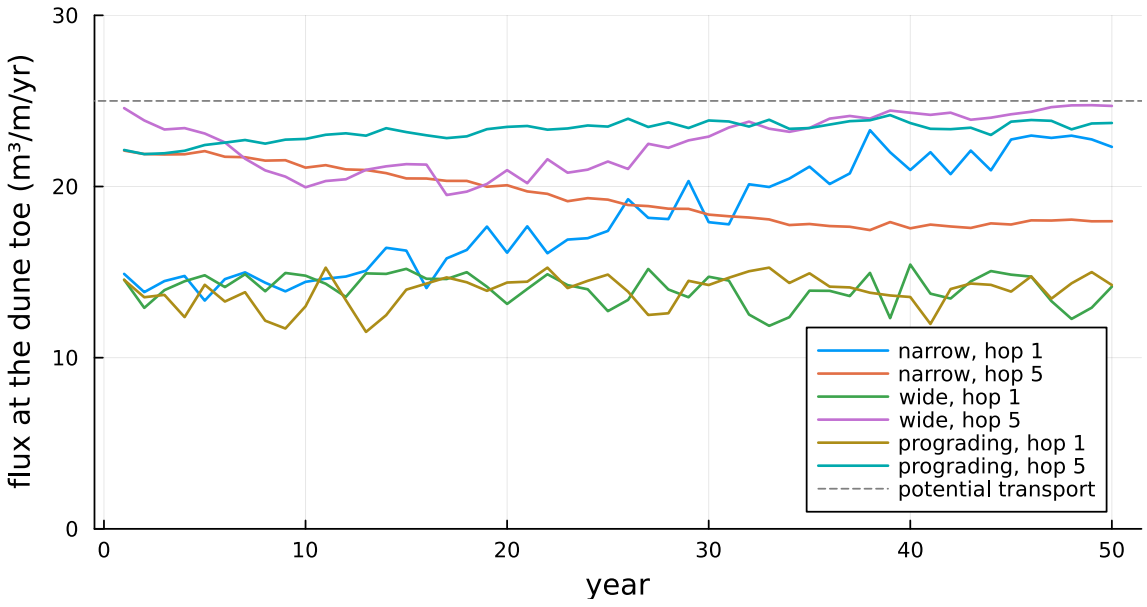

```
source_volume = Dict((:narrow, :default) => 1500, (:narrow, :saltation) => 1611,
                     (:wide, :default) => 1563, (:wide, :saltation) => 2234,
                     (:prograding, :default) => 1540, (:prograding, :saltation) =>
2130)
source_width = Dict((:narrow, :default) => 145, (:narrow, :saltation) => 150,
                    (:wide, :default) => 220, (:wide, :saltation) => 230,
                    (:prograding, :default) => 304, (:prograding, :saltation) =>
218)
[(beach=s, hop=rep === :saltation ? 5 : 1,
  toe_flux=round(mean(runs[(s, rep)].flux[2:end]); digits=1),
  volume_gain=round(Int,
                    runs[(s, rep)].measures[end].volume - runs[(s,
rep)].measures[1].volume),
  source_gain=source_volume[(s, rep)] - 1150,
  width=round(Int, runs[(s, rep)].measures[end].width),
source_width=source_width[(s, rep)])
 for (s, rep) in cases]
```

```
6-element Vector{@NamedTuple{beach::Symbol, hop::Int64, toe_flux::Float64,
volume_gain::Int64, source_gain::Int64, width::Int64, source_width::Int64}}:
 (beach = :narrow, hop = 1, toe_flux = 18.2, volume_gain = 371, source_gain = 350,
 width = 166, source_width = 145)
```

```
(beach = :narrow, hop = 5, toe_flux = 19.4, volume_gain = 572, source_gain = 461,
width = 159, source_width = 150)
(beach = :wide, hop = 1, toe_flux = 14.0, volume_gain = 392, source_gain = 413,
width = 235, source_width = 220)
(beach = :wide, hop = 5, toe_flux = 22.5, volume_gain = 1029, source_gain = 1084,
width = 242, source_width = 230)
(beach = :prograding, hop = 1, toe_flux = 13.9, volume_gain = 376, source_gain =
390, width = 274, source_width = 304)
(beach = :prograding, hop = 5, toe_flux = 23.3, volume_gain = 1045, source_gain =
980, width = 216, source_width = 218)
```

The sand crossing the dune toe, in m^3 per metre a year, the gain in dune volume over 50 years and the dune's width after them, against the source's medians (from an initial volume of about 1150 m^3/m; the medians of the narrow and wide beaches are in its text, those of the prograding beach and the wide beach's widths with hops of 1 cell read from its Fig. 4). As in the source, hops of 5 cells carry sand across a wide beach to the dune close to the potential transport, while single-cell hops build bedforms on the beach that hold sand back. So the dune gains much more with hops of 5 cells on the wide beach, here as in the source; on the narrow beach the fluxes differ less.

```
plot(1:50, [runs[c].flux for c in cases]; lw=1.5,
     label=permutedims(["$(c[1]), hop $(c[2] === :saltation ? 5 : 1)" for c in
cases]),
     xlabel="year", ylabel="flux at the dune toe (m³/m/yr)", ylims=(0, 30),
     size=(650, 340), legend=:bottomright)
hline!([25]; color=:gray, ls=:dash, label="potential transport")
```



On the prograding beach, with hops of one cell, the beach builds out faster than the dune can follow and smaller foredunes form on it, seaward of the main dune, which is wide, as in the source (274 m in this run, against its 304 m, its widest). With hops of 5 cells the sand reaches the

old dune, which grows a wide new ridge in front of it (216 m against 218 m), and the flux at the dune toe rises a little from the first decade (22.4 to about 23.6 $m^3/m/yr$) as the beach widens, as in the source's Fig. 5c.

Those are single runs, with means over rows. Over five seeds, with the source's statistic, the median over all profiles (and in brackets the range of the seeds' medians):

beach	hops	width (m)	source	gain (m^3/m)	source
narrow	1	166 (164-166)	145	370	350
narrow	5	163 (159-165)	150	583	461
wide	1	238.5 (228-245)	220	392	413
wide	5	258 (242-259)	230	1036	1084
prograding	1	284 (242-288)	304	380	390
prograding	5	213 (208-216)	218	1051	980

The gains are within about 7% of the source's, except on the narrow beach with hops of 5 cells, a quarter too large. The widths are off by 17.6 m on average: on the two beaches with a fixed shoreline the dune is too wide, by 13 to 21 m on the narrow and 18.5 and 28 m on the wide. An earlier version of this model reset a whole column at once and had a simpler plant rule. It gave 231 m on the wide beach with hops of 5 cells, but those rules are not the source's, so that agreement says little.

The narrow beach's dune advances at about a metre a year to the end of the run (127, 141, 151, 159 and 166 m by decade, with hops of 1 cell; seed 1). Its front is a nearly bare face at the angle of repose, rising 0.5 m per cell from about 1.3 m, with plant cover below a quarter up to about 7 m, its foot cut by the sea. The sea's strength does not set the rate: with no resistance from plants (`Rv = 0`) the width is still 166 m, with `G` of 0.7 or 0.9 it is 166 or 165 m, and with no dissipation (`Wdiss = 0`) 161 m. Each of these changes it (one run each, against 145 m and +350 m^3/m in the source):

change	width (m)	gain (m^3/m)
the run above	166	369
water levels 0.5 m higher	130	320
water levels 0.5 m lower	181	434
waves at their least strength everywhere	180.5	411
half the erosion probability	144	234
twice the deposition probability	141	242
no plants establishing anew	152	461
no new plants at all	144	368

The last row is close to the source's pair, though the source's plants do colonise, so it is no explanation. The source does not publish the probability that plants establish, 0.05 here from DuBeVeg's code. In a sweep from 0 to 0.1 over the six cases, run on this model before its avalanches took their present form, no value fitted both the widths and the gains: lower values

narrowed the dunes on the fixed shorelines and made the narrow beach gain far too much. Nor does it publish how the dissipation `D` accumulates. The waves follow DuBeVeg's code: their initial strength is the total water level (in slabs) times an energy factor of 1, which keeps them strong far up the beach and, near the shoreline, overrules the protection of plants; the source's Eq. 4 leaves the factor out. Vignette 28 shows that a wind steered by the topography also narrows the dunes, though the source's wind is not steered.

5 Deviations from the source

```
foreach(d -> println("- ", d), model_info(:dune_formation).deviations)
```

```
- The sea is forced by the water levels measured at Scheveningen (1970–2020), the
highest in each spring–neap cycle, as in the source, taken from Rijkswaterstaat
rather than from the source, which only plots them. They are against NAP, where the
source writes mean sea level, with no correction. A synthetic series with much the
same distribution (1.43 m plus a Gumbel-distributed surge of scale 0.24 m) remains
available with levels = :synthetic.
- The domain is 20 to 100 rows of 1 m alongshore (400 in the source) and only the
cross-shore profile from 60 m offshore to 20 m behind the dune (1250 cells in the
source). The initial profile is a Dean-like shoreface, a linear beach and a smooth
dune ridge to 15 m read from the source's figure; the beach rises to 2.5 m at the
dune foot, as the figure shows, where the text says 3 m (with 3 m, small bumps on
the upper beach count as the dune).
- Slab transport is synchronous, as in DuBeVeg's code: all pick-ups are drawn from
the state at the start of an iteration, with the shadow and the probabilities fixed
for its duration.
- Part-way through a hop (Eq. 5), a slab lands only where vegetation is at least
rho_star, and the shadow does not force it to; after a failed landing at a hop's end
the next hop begins. Slabs reaching the land side of the domain are lost.
- The shadow test uses the continuous threshold i tan(15°); on whole slabs it is the
same test as DuBeVeg's floored one, except that DuBeVeg's code looks upwind no
further than the highest elevation divided by tan(15°), which matters only for cells
below zero.
- Constants not in the source come from DuBeVeg: the growth functions of the pioneer
(burial -1.4, 0.2, 0.6, 2.0, 2.2 m/yr, peak 0.2) and stabilizer (-1.4, -0.65, 0,
0.2, 2.8, peak 0.05), lateral expansion 0.2 and establishment 0.05 a year (each
species spreading only from neighbours of its own, its yearly gain at most the cover
left free, establishment only from 0 to 20 m high, where the source gives every bare
cell the same chance, and for new stabilizers only where the sand balance is zero,
where DuBeVeg's code asks that no sand moved at all that year), the wave run-up
relation, the dissipation by the inverse water depth summed from the sea, the least
wave probability 0.1, and the waves' initial strength, the total water level in
slabs times Fenergy = 1 (wave_energy), which multiplies WE - Wdiss D in DuBeVeg's
code (the source's Eq. 4 leaves it out); near the shoreline it overrules the
protection of vegetation, as in DuBeVeg. Plant cover is multiplied each year by one
minus the sum of the sea's probabilities over the year (at most 1), as in DuBeVeg's
code, and otherwise stays through each reset. The initial split of species is not
stated: pioneers seaward of the crest, stabilizers behind.
```

- Avalanching follows DuBeVeg's code: each cell passes a slab to its steepest lower neighbour (by height difference) only if that drop exceeds the angle of repose (times the square root of 2 along a diagonal). The source allows only slopes equal to or gentler than the angle of repose; the code leaves an orthogonal slope steeper than that where a lower diagonal neighbour is within its own limit.
- The dune is wider than in the source after 50 years, medians of five seeds, on the narrow beach (166 and 163 m with hops of 1 and 5 cells, against 145 and 150) and on the wide beach (238.5 and 258 m against 220 and 230), and the narrow beach with hops of 5 cells gains too much (583 m^3/m against 461). The cause is not known. On the narrow beach the front is a nearly bare face at the angle of repose, its foot cut by the sea, advancing about a metre a year, and the sea's strength does not set the rate. With no new plants at all the narrow beach with hops of 1 cell gives 144 m and +368 m^3/m (the source 145 and +350), but the source's plants do colonise; it does not publish the probability that they establish, nor how the dissipation D accumulates.
- The prograding beach builds out through the sea alone: the reference surface is replaced at the end of each year by the profile with a berm 4 m longer, the profile below the berm moved seaward within a domain extending 200 m further out (the source's profile reaches a depth of closure of 8 m, beyond the domain here), and the sea's resets move the beach towards it; the shoreline then follows the reference about a year behind.
- The sensitivity to rho_star is not included; one run per case on 20 rows replaces the source's five runs on 400.
- Beyond the source: three more landforms built from the same profile pieces (an embryo dune and slack seaward of the foredune on the wide beach, a seawall and promenade at the back of the narrow beach, and a breach through the foredune in a fifth of the rows), and a non-erodible layer (hard) for the seawall; the source runs none of them, so they test the model's behaviour, not its fidelity. A breach fills from its flanks rather than growing into a blowout.
- Beyond the source: an optional wind steered by the topography (speedup > 0; off by default), from the linear theory of turbulent flow over low hills (Kroy et al. 2002 along the wind, Schwammle and Herrmann 2005 across it), computed on the height above the reference surface with the shadow envelope in the lee. The erosion probability scales with the shear along the wind and flying slabs turn alongshore with the shear across it. With A = 4 it narrows the dunes, a little past the source's width on both beaches with hops of 5 cells (144 and 219 m against 150 and 230), but a breach still fills.

Teixeira, Manuel, Erik M. Horstman, and Kathelijne M. Wijnberg. 2023. “Conceptualizing Aeolian Sediment Transport in a Cellular Automata Model to Simulate the Bio-Geomorphological Evolution of Beach–Dune Systems.” *Journal of Marine Science and Engineering* 11 (7): 1278. https://doi.org/10.3390/jmse11071278.